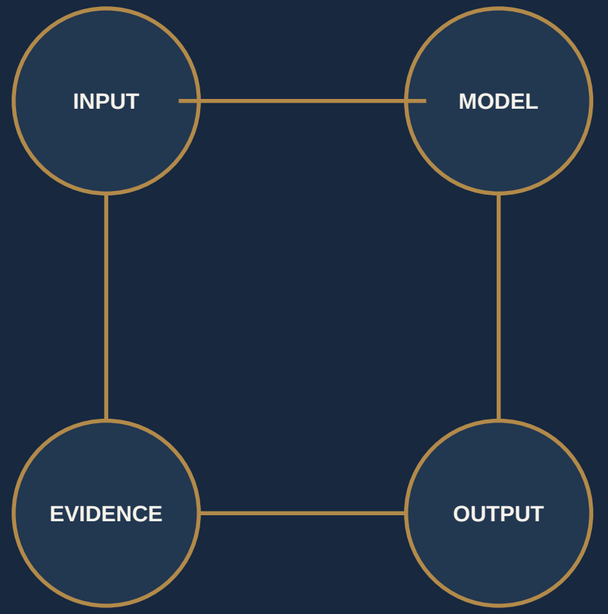
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Embedded Systems

AI.109 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define microcontrollers.
- Apply key methods including processors.
- Build or analyze a practical system involving memory.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Microcontrollers

02

METHODS

Processors

03

SYSTEMS

Memory

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Microcontrollers
Microcontrollers

02 · Processors
Processors

03 · Memory
Memory

04 · GPIO
GPIO

05 · Timers
Timers

06 · Interrupts
Interrupts

07 · Communication protocols
Communication protocols

08 · Sensors
Sensors

Core ideas, made explicit

AI.109 • Distinguish definitions, assumptions and evidence

- Microcontrollers
- Processors
- Memory
- GPIO
- Timers

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.109 • Distinguish definitions, assumptions and evidence

- Interrupts
- Communication protocols
- Sensors
- Actuators
- Real-time systems and embedded programming

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Microcontrollers

INPUTS
Processors

METHOD
Memory

OUTPUT
GPIO

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving microcontrollers.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using microcontrollers.
- Implement or simulate a baseline using processors.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.